Barts Cancer Institute
Queen Mary University of London

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MSc Computational Biology
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Proteomics and Phosphoproteomics

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Mass Spectrometry Facility
Barts Cancer Institute

CANCER RESEARCH UK | BARTS CENTRE

Learning objectives

- To understand the rationale for using proteomics to complement genomic approaches
- To familiarize with proteomics technologies
- To be aware of other “omics” technologies related to proteomics (phosphoproteomics)
- Introduce computational methods for the analsyis of proteomics and phosphoproteomics data

Session plan

1. Lecture

2. Dry lab practical:
   - Dry lab 1 – Peptide and phosphopeptide identification from mass spectrometry data
   - Dry lab 2 – Statistical analysis of quantitate phosphoproteomics data
   - Dry lab 3 – Kinase activity estimation from phosphoproteomics

By the end of today you will be able to:

- Identify proteins from mass spectrometry data
- Carry out differential analysis of proteomics and phosphoproteomics data
- Estimate changes in protein kinase activity from phosphoproteomics data

Some definitions

- Proteome: the entire complement of proteins in a given biological system such as cell type, biological fluid or tissue.
  - Dynamic: a living being has just one genome but many proteomes (one per each cell type) and these change during development and as a result of environmental factors (e.g., fed versus starved conditions and in disease).
- Proteomics: technology to study the proteome. Analysis of <u>prote</u>ins in a gen<u>omics</u> scale.

The central dogma of biology

The central dogma in biology

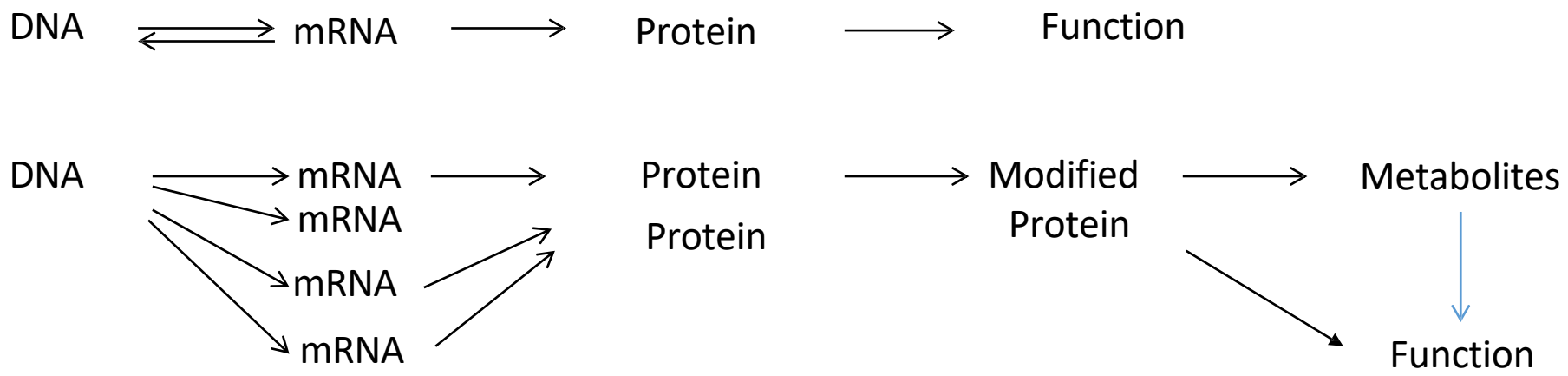
DNA $\rightleftharpoons$ mRNA $\longrightarrow$ Protein $\longrightarrow$ Function

Transcription Translation Enzyme activity

One gene = One function

Reality is more complex

The central dogma in biology



gene

Not all genes expressed equally. Different transcripts (splice variants)

Not all transcripts translated into proteins

Not all proteins are active (PTMs, localization)

genomics | transcriptomics | proteomics | phosphoproteomics | Metabolomics

Proteomics

- Provides biological information not attainable with genomic technology.
- Aims to identify and characterize protein alterations linked to disease because proteins can be:
  - Drug targets
  - Biomarkers
- Involves identifying proteins and comparing them across different sample groups or conditions.
- Emphasizes an unbiased and untargeted workflow to ensure a thorough coverage of the proteome.

Examples of applications

- Identification of biomarkers of disease in tissues and bio-fluids
  - Discovery of biomarkers for early detection of cancer
- Identification of biomarkers of responses to cancer drugs for personalized medicine
  - Determine which cancer drug should be given to a cancer patient
- Understanding the mechanisms governing normal and pathological biological processes at a systems biology level.
  - Analysing changes in protein phosphorylation due to disease or treatments
  - Tracking protein movement across cellular organelles
  - Monitoring changes of protein expression and protein synthesis dynamics
  - Studying protein-protein interactions
- Elucidating the mechanisms of action of cancer drugs
  - Assessing alterations in protein abundance and modifications following drug treatments

Technological challenges

- Chemistry of proteins is more complex than that of nucleic acids
  - DNA has only four bases with similar chemistry : acidic (-ve charged) sugars
  - 20 different a.a. with large differences in chemistry:
    - Polar, Hydrophobic, basis, acidic:
- Two copies per gene; large dynamic range of protein expression in cells and bio-fluids.
- Proteins cannot be amplified experimentally, in contrast to DNA (with PCR)

Proteomics technology

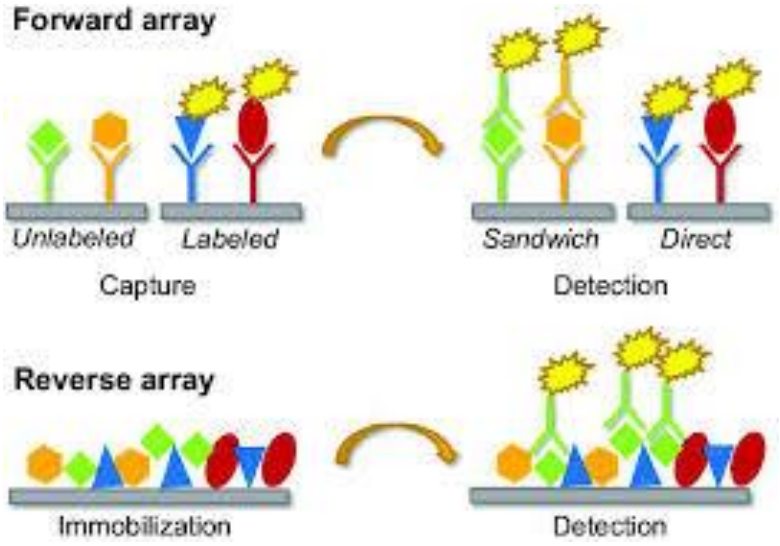
- Targeted analysis
  - Needs preconception of proteins that could be involved in the question being addressed.
  - Normally limited to the detection of a few hundred proteins.
  - Antibody / Mass spectrometry based
- Untargeted analysis
  - No need of preconception of proteins that could be involved in the question being addressed.
  - Mass spectrometry based

Separation / Identification / Quantification

- LC-MS/MS
- Gel followed by MS/MS or LC-MS/MS (Gel used for Separation or Separation & Quantification)

Targeted proteomics: protein arrays

Antibody Based:



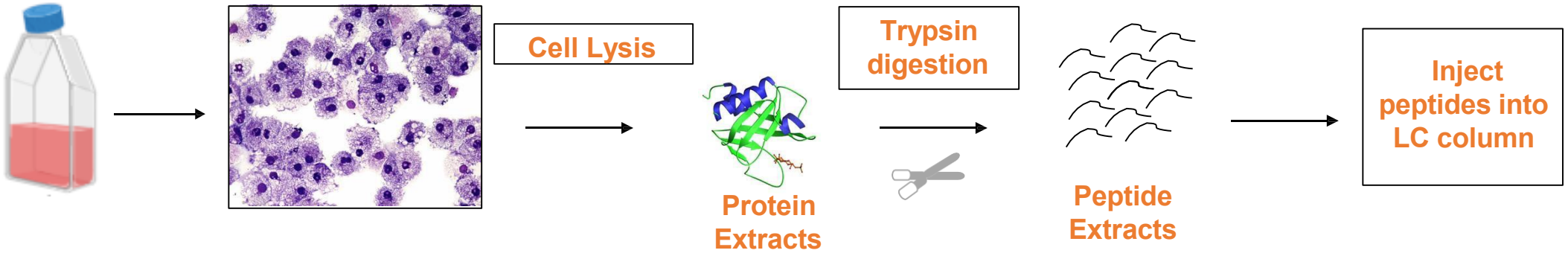
•DOI: 10.2147/ANTI.S53335

Type	Immobilized Molecule	Detection Strategy	Measurement
Forward Sandwich	Antibodies	Labelled Antibody	Multiple Proteins / Few Samples
Forward Direct	Antibodies	Labelled Proteins	Multiple Proteins / Few Samples
Reverse	Proteins	Labelled Antibody	Single Protein / Many Samples.

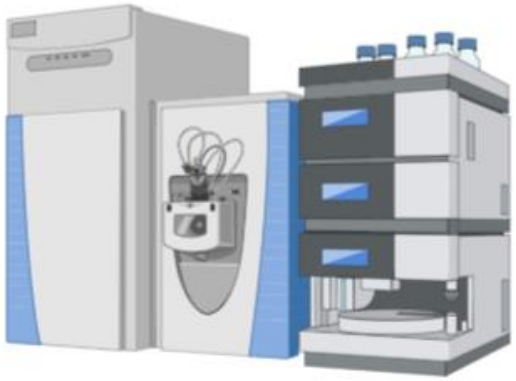
- Restricted to antibody availability.

Untargeted proteomics by tandem mass spectrometry (MS/MS)

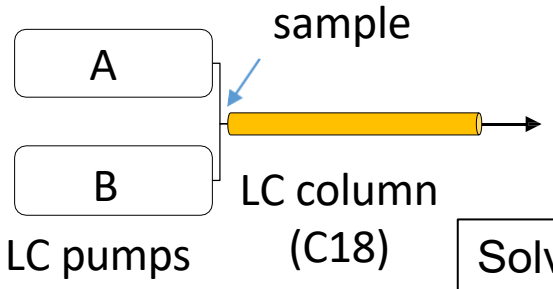
A Sample Preparation



B LC-MS/MS



Column in LC separates peptides based on hydrophobicity

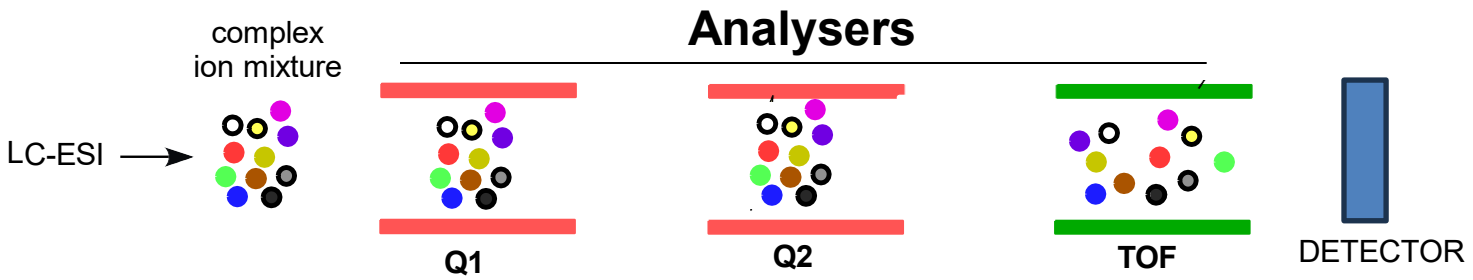


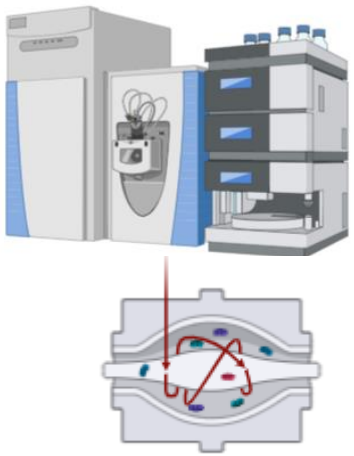
As peptides elute from LC column, these are analysed by tandem mass spectrometry (MS/MS). Peptides are sequenced and "quantified".

Solvent A: H2O + 0.1% FA
Solvent B: ACN + 0.1% FA

Glossary
LC, liquid chromatography: column filled with material to which peptides bind to. Changing the flow of solvent from aqueous to organic elutes peptides from column in order or hydrophobicity (stickiness to column material)
MS/MS, tandem mass spectrometry: an instrument that measures the mass of compounds, then fragment such compounds to reveal their structures; in the case of peptides this is the primary structure (sequence).

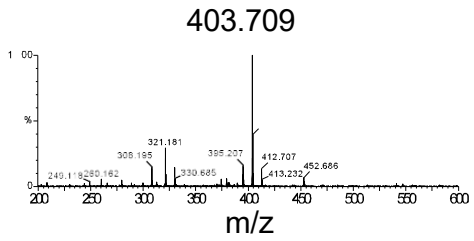
Untargeted proteomics by tandem mass spectrometry (DDA)





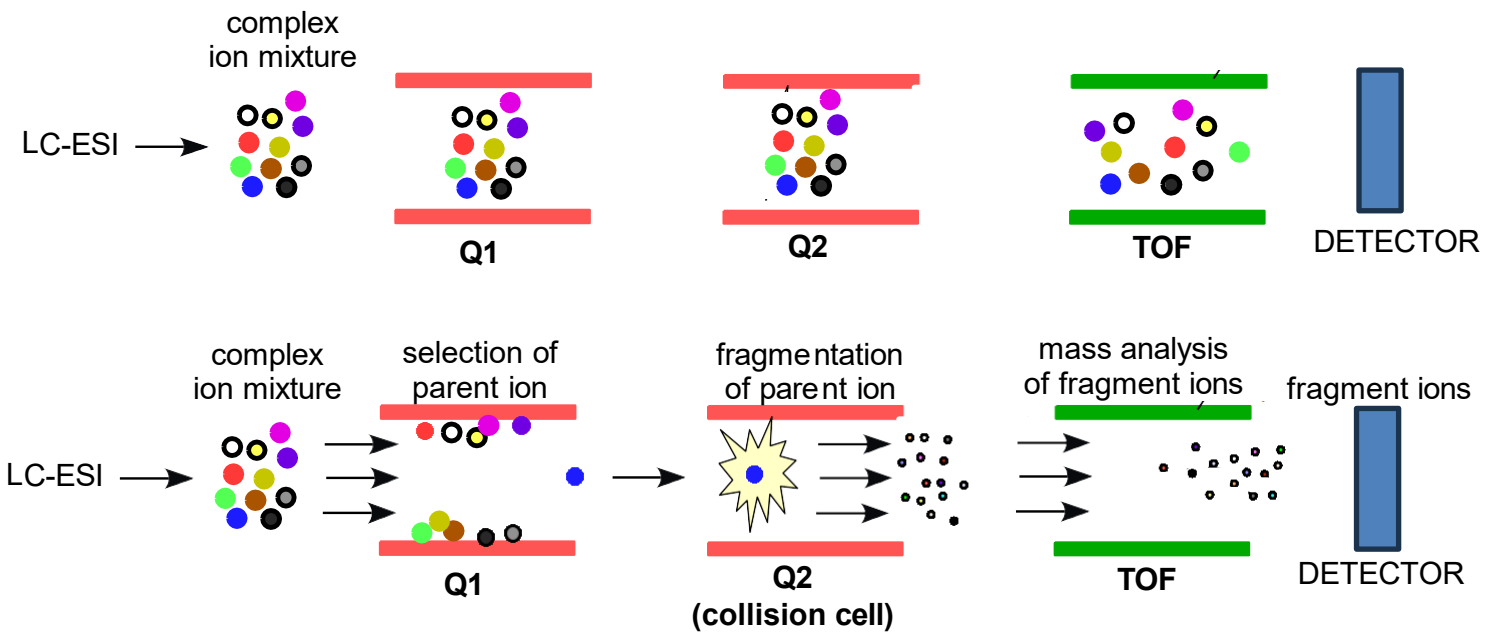
Orbitrap
Analyser

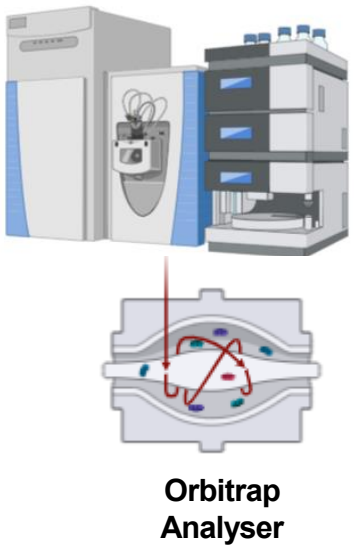
Informatics
Peptide
sequences



DDA: Data Dependent Acquisition

Untargeted proteomics by tandem mass spectrometry (DDA)





Informatics
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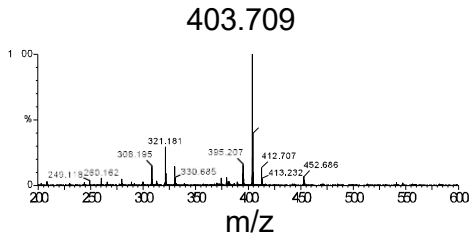

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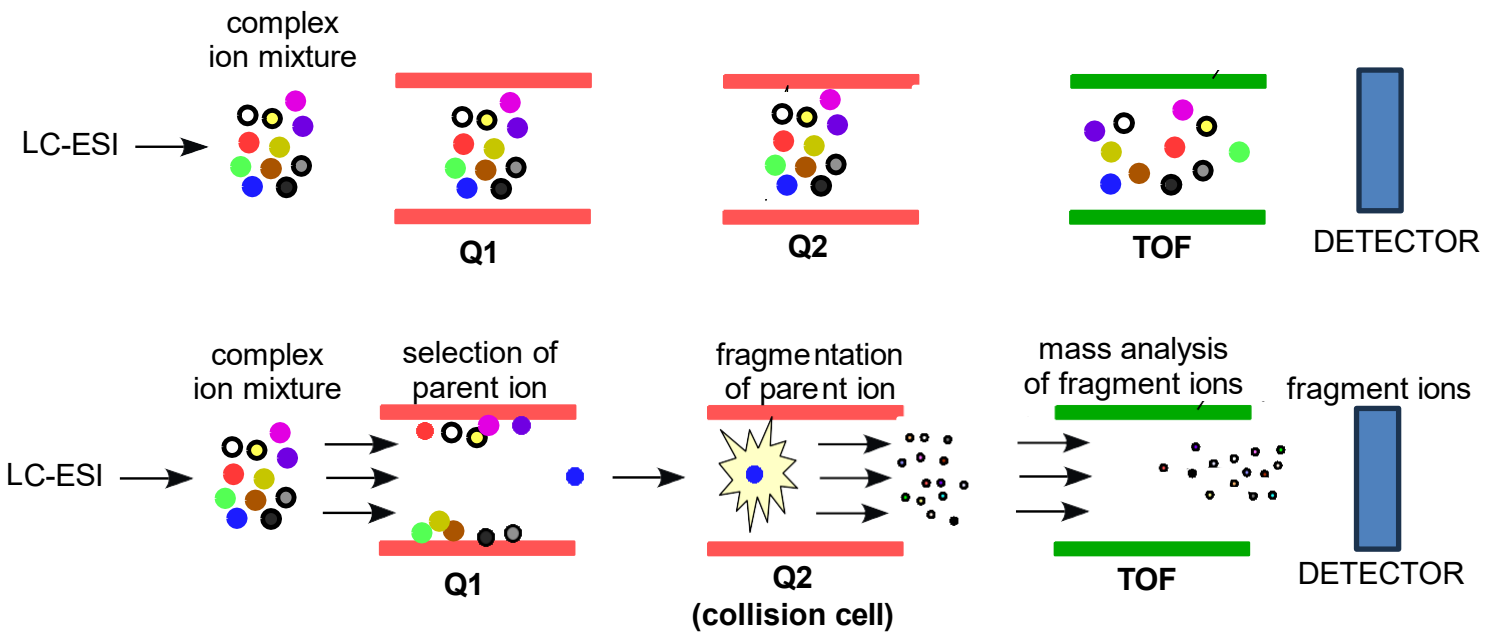

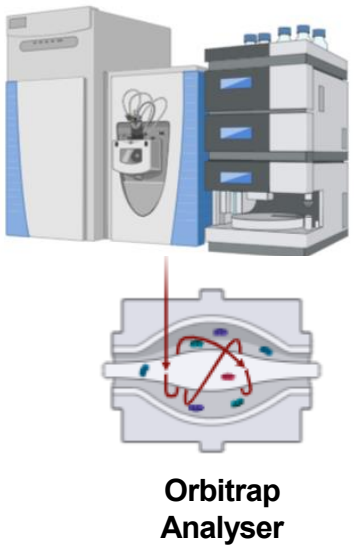

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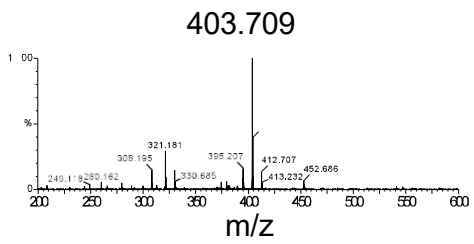

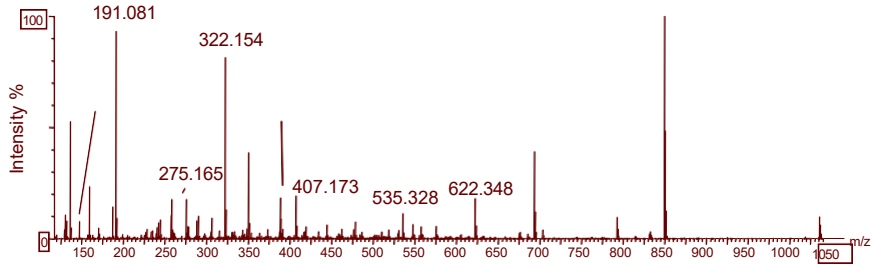

Image from: Nature Reviews Molecular Cell Biology **5**, 699-711 (September 2004) doi:10.1038/nrm1468

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Peptide fragmentation patterns (MS/MS spectra) are used to infer peptide sequence

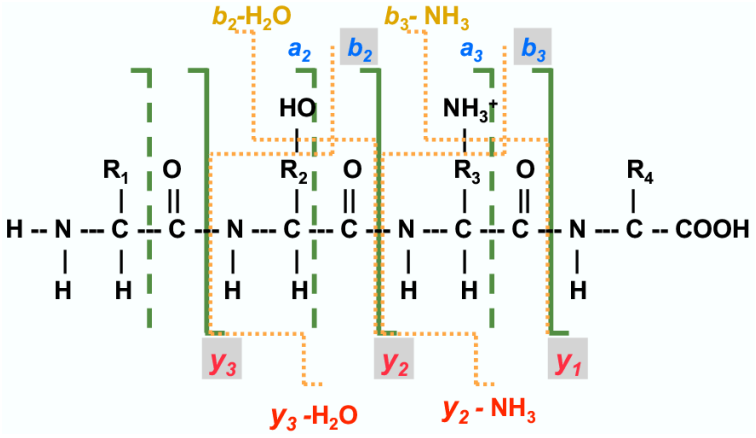


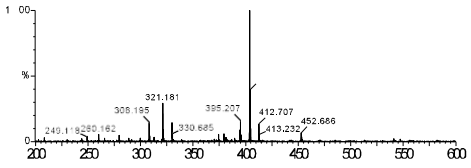
Table 7.2 Mass increments of the various amino acids

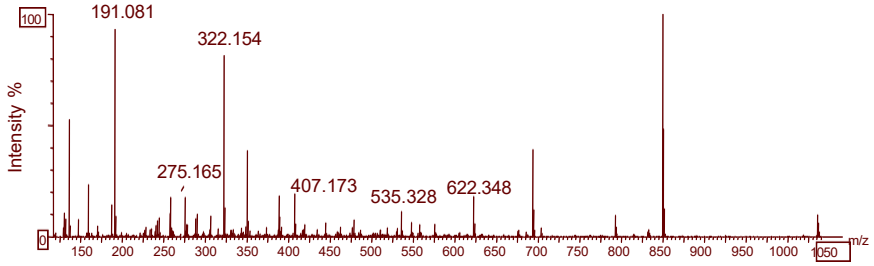
Amino acid	Code (3 letters)	Code (1 letter)	Monoisotopic mass	Chemical mass
Glycine	Gly	G	57.02147	57.052
Alanine	Ala	A	71.03712	71.079
Serine	Ser	S	87.03203	87.078
Proline	Pro	P	97.05277	97.117
Valine	Val	V	99.06842	99.133
Threonine	Thr	T	101.04768	101.105
Cysteine	Cys	C	103.00919	103.144
Isoleucine	Ile	I	113.08407	113.160
Leucine	Leu	L	113.08407	113.160
Asparagine	Asn	N	114.04293	114.104
Aspartate	Asp	D	115.02695	115.089
Glutamine	Gln	Q	128.05858	128.131
Lysine	Lys	K	128.09497	128.174
Glutamate	Glu	E	129.04260	129.116
Methionine	Met	M	131.04049	131.198
Histidine	His	H	137.05891	137.142
Phenylalanine	Phe	F	147.06842	147.177
Arginine	Arg	R	156.10112	156.188
Tyrosine	Tyr	Y	163.06333	163.17
Tryptophan	Try	W	186.07932	186.213

Precursor:	PEPTIDE:	1800 (z2)
Fragments:	E (y1):	129 (z1),
	DE (y2):	244 (z1)
	IDE (y3):	357 (z1)
	TIDE (y4):	458 (z1)
	PTIDE (y5):	555 (z1)
	EPTIDE(y6)	684 (z1)
	PEPTIDE (y7)	781 (z1)

Database

m/z
403.709





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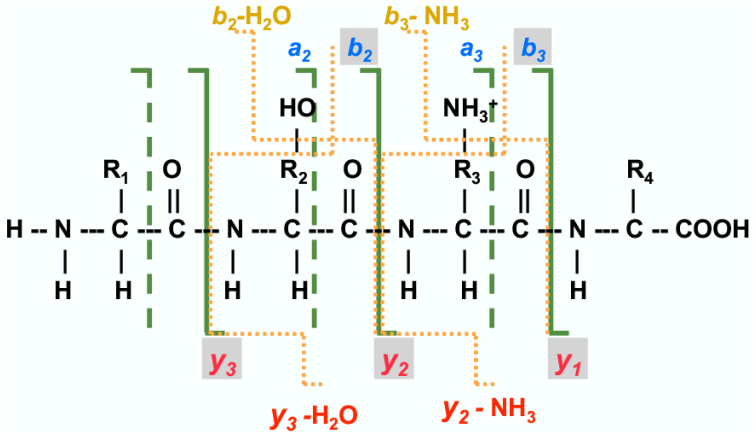

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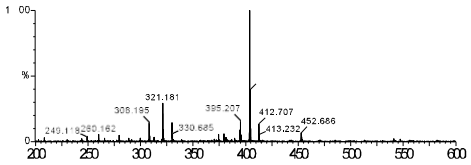

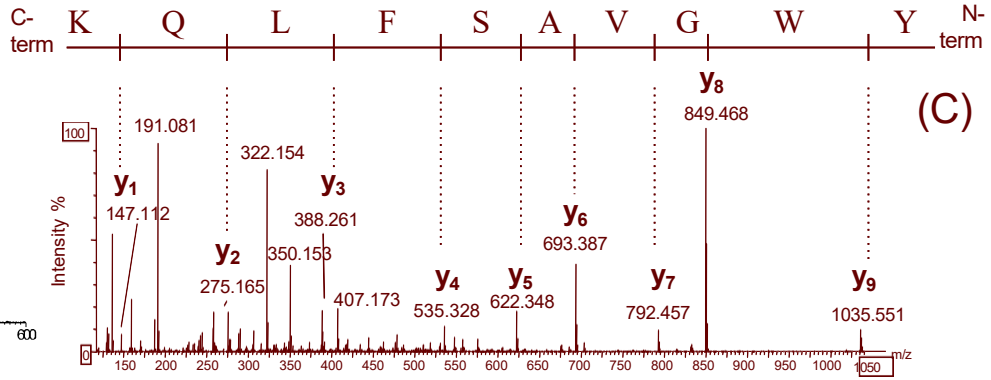

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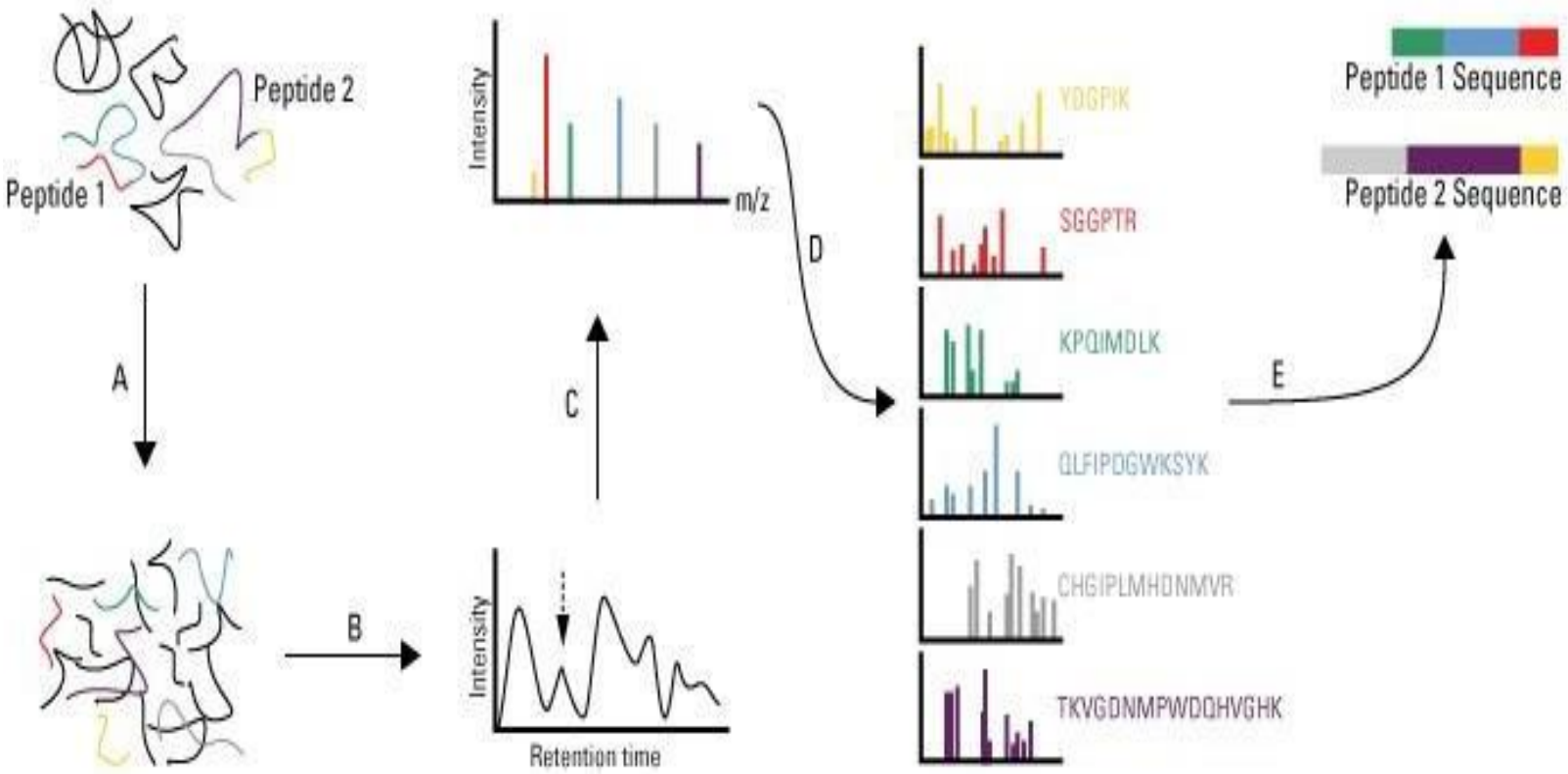
Untargeted proteomics by data dependent analysis (DDA)

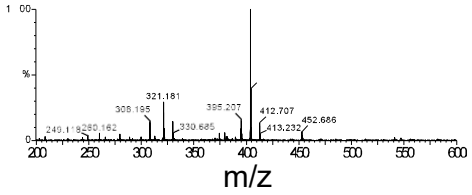


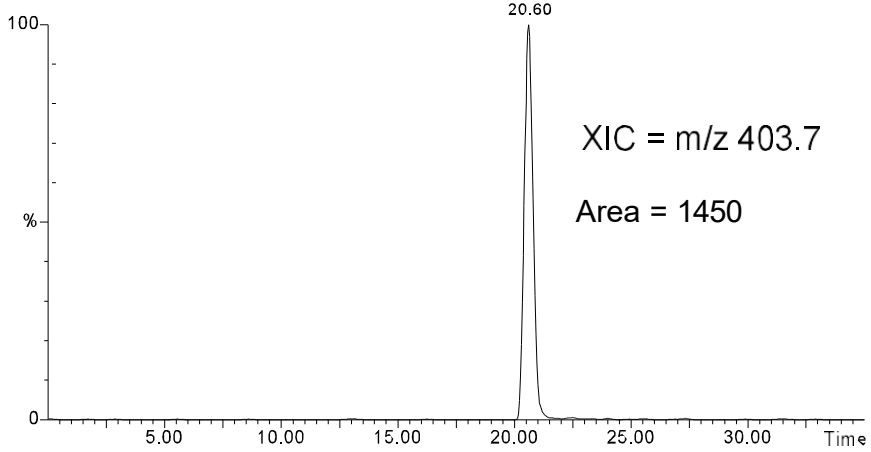
https://www.thermofisher.com/pierce-protein-methods/quantitative-proteomics.html

Untargeted proteomics by tandem mass spectrometry (MS/MS)

Analyte: YWGVASFLQK

403.709





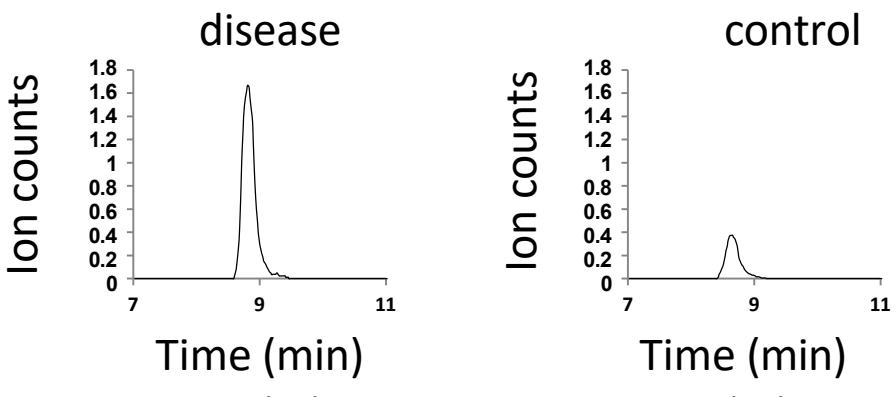
XIC = m/z 403.7

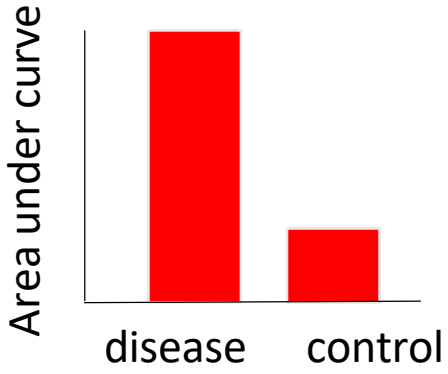
Area = 1450

Relative quantification to compare peptide/protein abundance

Label-free

Elution profile of peptide





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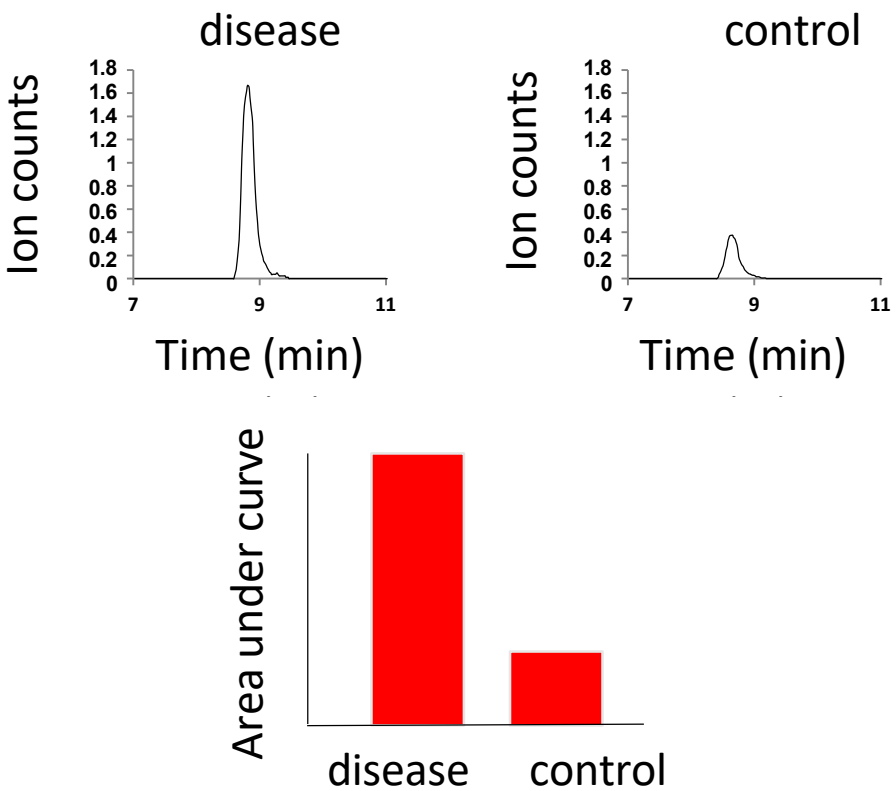
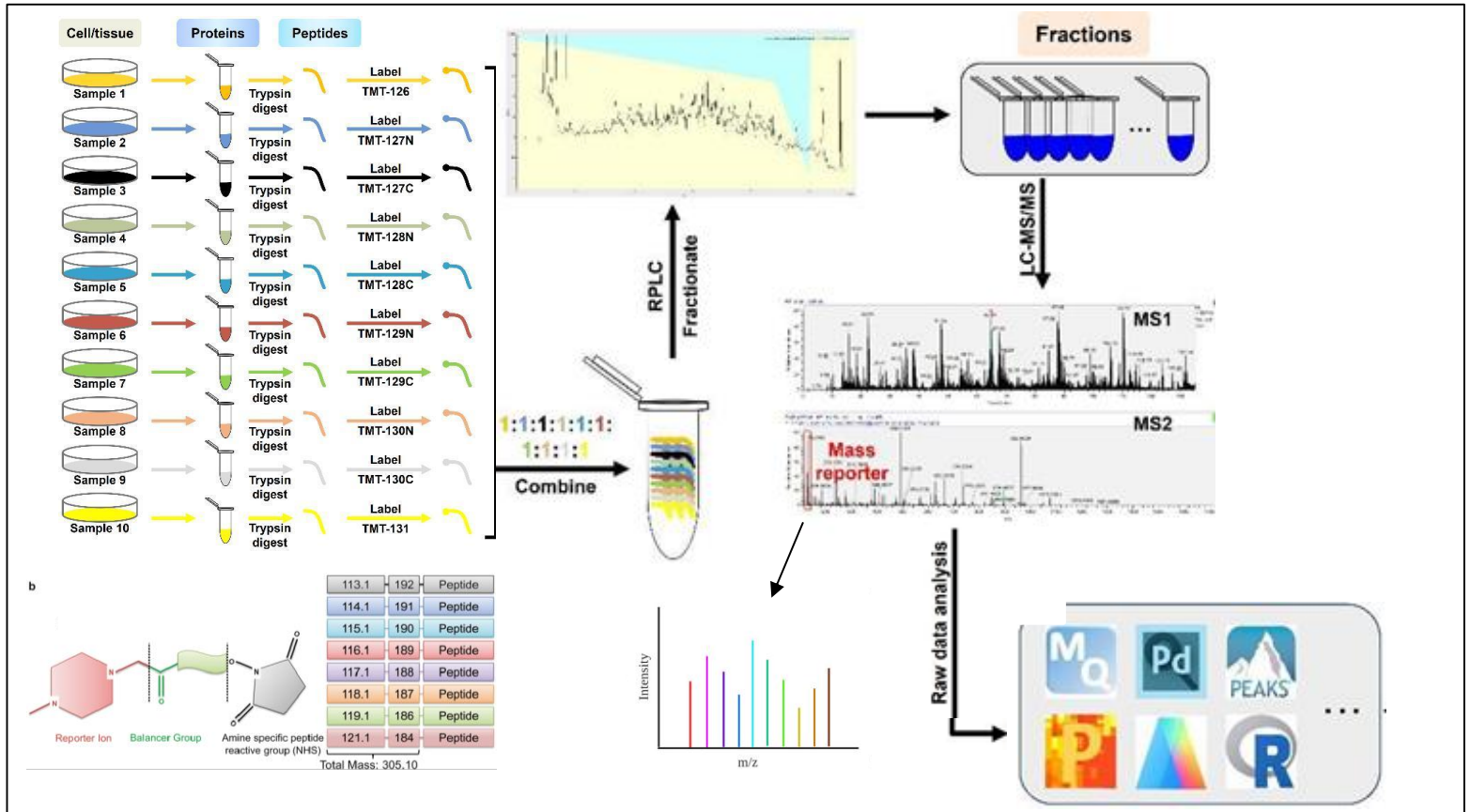


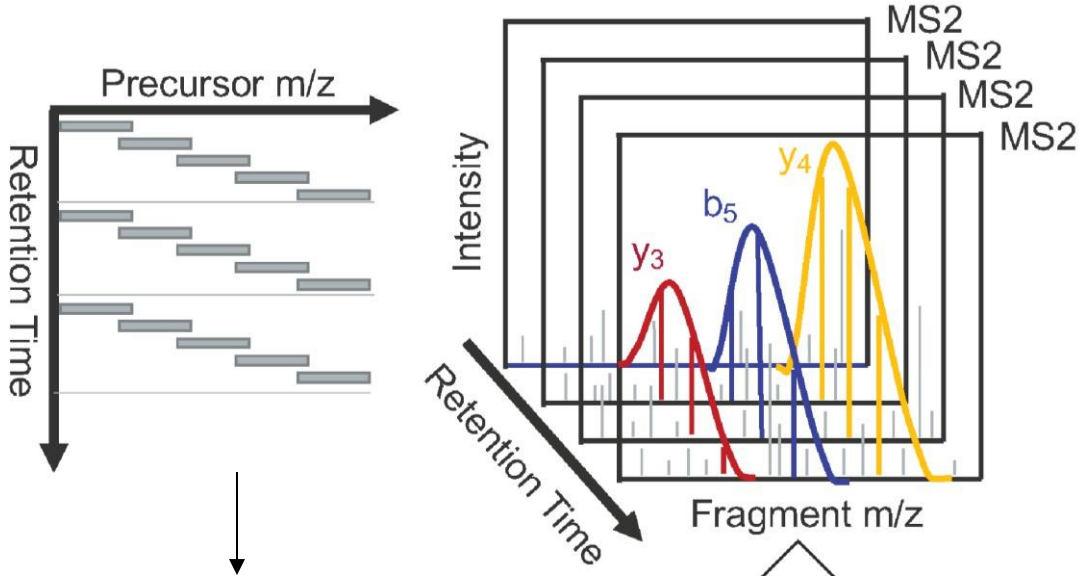
Label-based

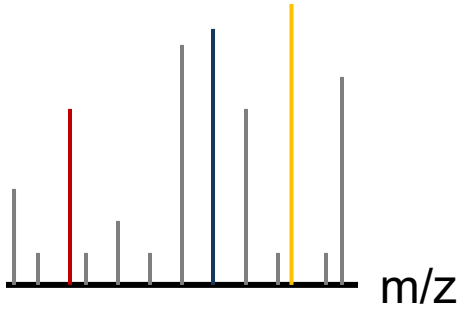
Metabolic (eg. SILAC), or chemical (e.g., TMT) labelling



https://www.creative-proteomics.com/services/tmt-based-proteomics-service.htm

Untargeted proteomics by data independent analysis (DIA)



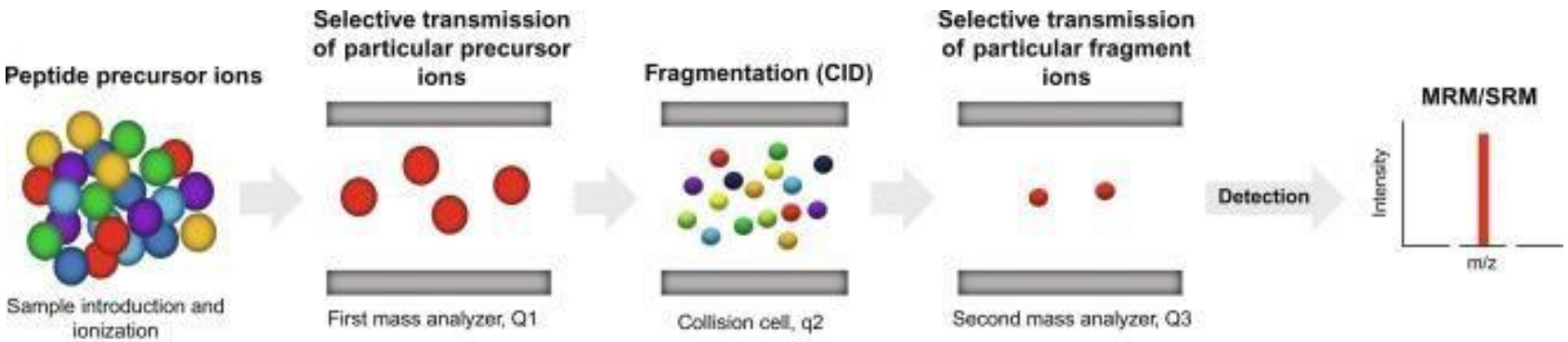


DIA: Data Independent Acquisition

https://www.openms.de/comp/diaproteomics/

Targeted proteomics using mass spectrometry

Selected reaction monitoring (SRM)



Parallel reaction monitoring (PRM)

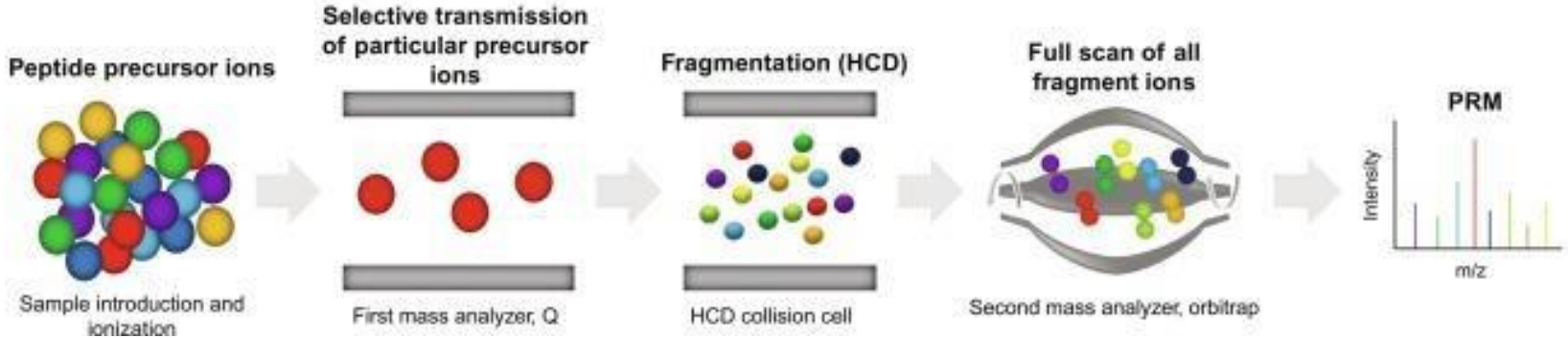


Image from: Proteomic and Metabolomic Approaches to Biomarker Discovery. Edited by: Haleem J. Issaq and Timothy D. Veenstra

Summary

- Global analysis of proteins (Proteomics) is more challenging than DNA sequencing - proteomics lags behind genomics in terms of technology
- LC-MS/MS technology developing fast

Motivation:

- Proteins are the ultimate mediators of biological function and the targets of most drugs
- mRNA levels do not always correlate with protein levels
- Need to consider proteins for a better understanding of cell biology and mechanisms of disease

However…

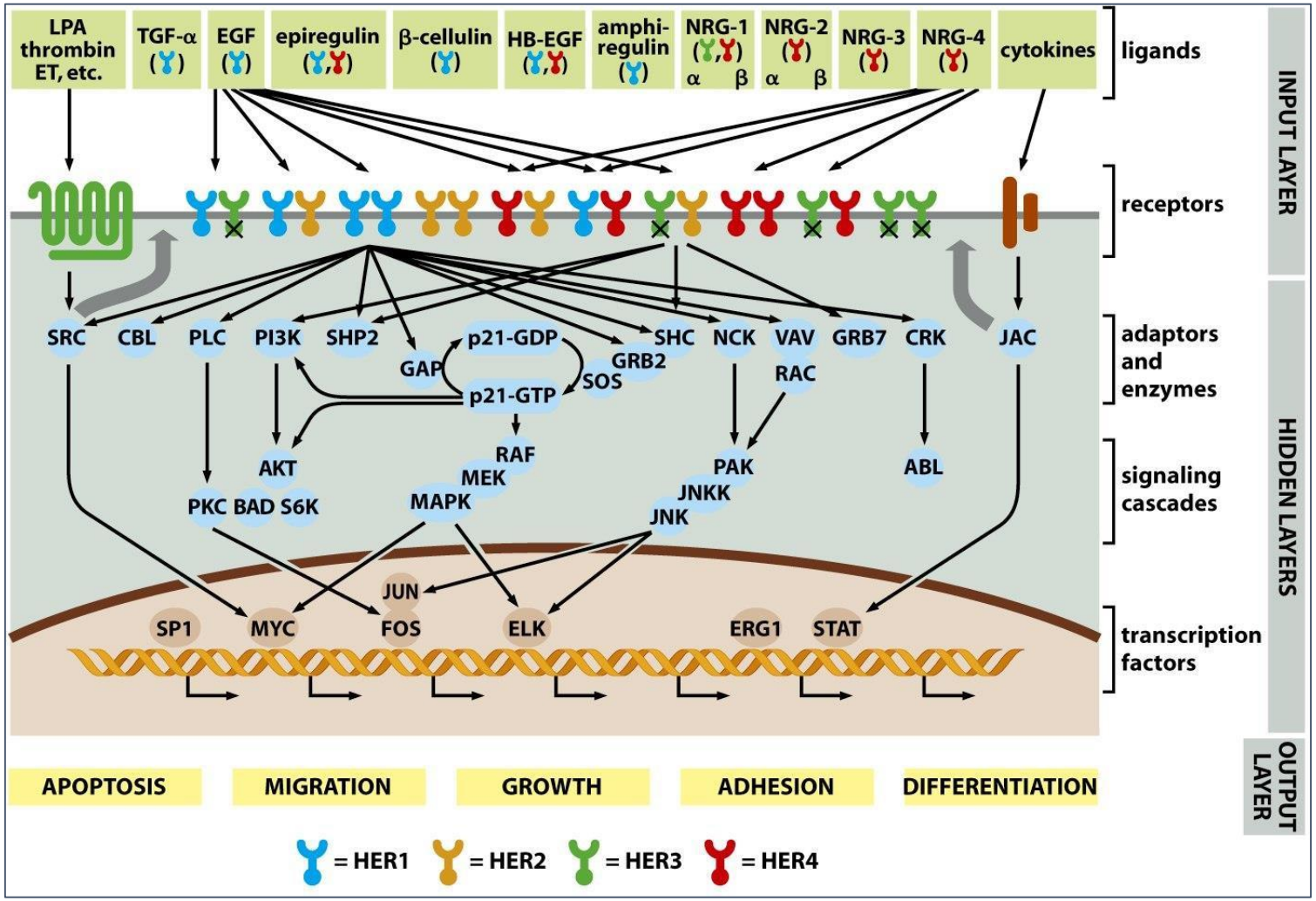
- Regulatory proteins and drug targets are enzymes
- Protein copy numbers in cells is only one factor in determining enzyme activity
- Protein (enzyme) copy numbers do not necessarily correlate with how active such enzyme may be in cells

Mechanisms of enzyme regulation

- Expression
- Covalent modifications, such as
  - Phosphorylation, Acetylation, Lipidation
- Allosteric regulation
  - e.g., feedback inhibition: end product of metabolic pathway inhibits enzyme that catalyses the rate limiting step in the pathway.
- Cofactors
  - Holoenzymes (inactive) may require cofactors to become active (apoenzyme).
- Proteolytic cleavage
  - E.g., some enzymes (zymogens) need to be cleaved to be active
- Protein-protein interactions
- Subcellular localization

Biochemistry, 5th edition
Jeremy M Berg, John L Tymoczko, and Lubert Stryer.

Example: cell signalling



Growth factors

↓

Receptors

↓

Signalling pathways

↓

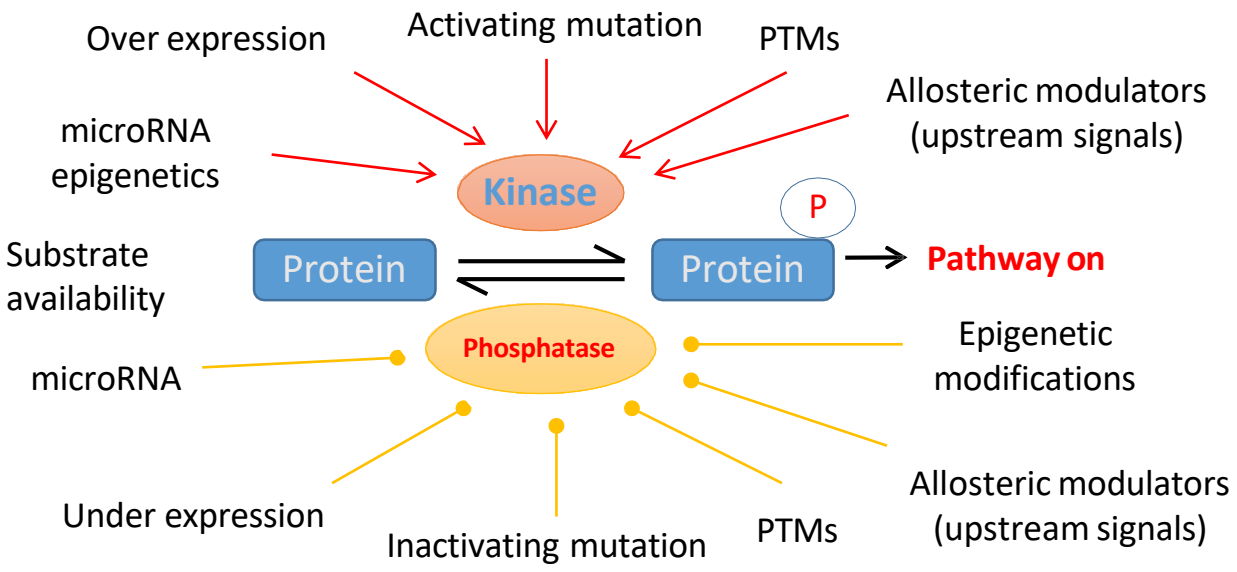
Aberrant activation

↓

Uncontrolled cell proliferation

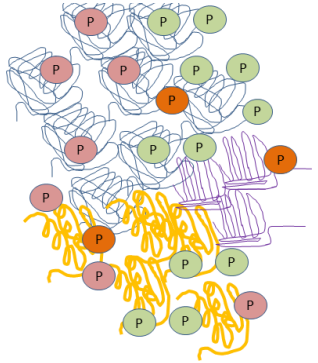
Figure 5.1 ***The Biology of Cancer*** **(© Garland Science 2013)**

Quantifying phosphoproteins as a means to infer kinase pathway activity



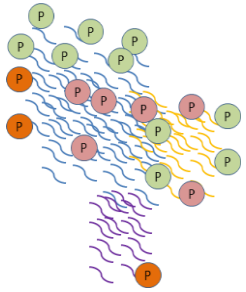
Phosphoproteomics data are used to quantify kinase activities on a global scale

Phosphoproteomics



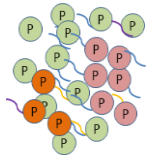
Proteins and phosphoproteins present in cell lysate.

trypsin →



peptides

IMAC/ TiO2 →



phosphopeptides

LC-MS/MS →

Single shot
Label-fee Quant

Overview of phosphoproteomic datasets

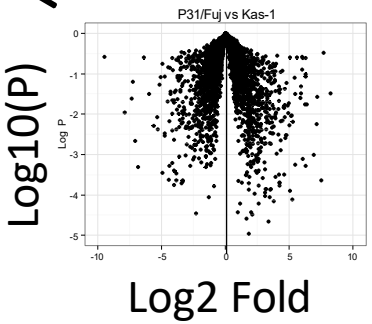
- Thousands (typically 10k-15k) of phosphorylated peptides (phosphopeptides) identified in a typical experiment
- Each phosphopeptide contains one or more phosphorylation site.
- Complex data from which we need to obtain biochemical information

Bioinformatic approaches to analyse proteomics and phosphoproteomics data

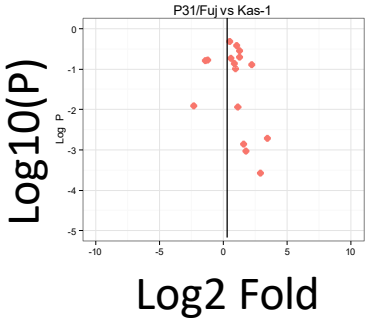
- Bioinformatic approaches are needed to derive biological information from proteomics and phosphoproteomics data
- Different methods available
- Essentially all of them involve performing statistical enrichment analysis of ontologies or pathways (for proteomics) or kinase-substrates (for phosphoproteomics)
- Examples:
  - Gene ontology enrichment analysis – proteomics
  - Kinase substrate enrichment analysis – phosphoproteomics
  - Motif analysis - phosphoproteomics

Example: Kinase substrate enrichment analysis

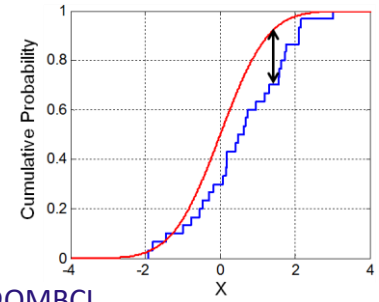
Increased in Kas-1 ← → Increased in P31/Fuj



All phosphorylation sites



Akt substrates



Parametric test

$$KSEA_{Akt} = \frac{(x - \mu) * sqr(m)}{\sigma}$$

x = median fold change of phosphorylated peptides that are substrates of a given kinase (Akt in the example)

μ = median fold change of all phosphorylated peptides

δ = standard deviation of fold change of all phosphopeptides

m = total number of phosphopeptides known to be Akt substrates in the dataset

Nonparametric test

P-value = ks.test(α,β)

Comparison of cumulative distributions by Kolmogorov-Smirnov test

α = vector of all phosphopeptide fold changes

β = vector of phosphopeptide fold changes of substrates of a given kinase

Illustration of cumulative probability from: https://commons.wikimedia.org/w/index.php?curid=25222928

General bioinformatic pipeline for DDA proteomics

Software	Tasks	File formats	
Vendor specific	Mass spectrometry raw data	.raw, mzML	Lab 1
Mascot, Andromeda, MSFragger, Protein Prospector, Open MS	Peak list	.mgf, .pkl, mzDATA	
	Peptide ids	.csv, .DAT, mzIdentML	
MaxQuant, Pescal, XLMS, others	Quantitative values	.csv, .txt	
Functions in R, Python, MS vendor software	Normalised Quantitative values		Lab 2
Limma, t-test, Perseus	Statistical analysis of differential abundance		
GSEA, KSEA, DAVID bioinformatics	Biological interpretation / visualization		Lab 3

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